JunctionIQ

SMART TRAFFIC MANAGEMENT SYSTEM FOR URBAN CONGESTION CONTROL



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PROBLEM:

–: Problem Statement :–

Smart Traffic Management System for Urban Congestion Control.

SIH 25050 - Software edition - Government of Odisha.

Traffic signals often operate on predetermined timings that may not reflect the actual number of vehicles waiting in each lane.

1. **Fixed Timers:** Traffic lights run on a set countdown, changing colors even if no one is there.
2. **Long waiting times:** Vehicles wait even when traffic demand differs between lanes.
3. **Emergency delays:** Congestion can delay emergency vehicles travelling through intersections.
4. **Blind System:** Because the lights cannot "see" the cars, you often get stuck waiting at a red light while the green light is given to a completely empty road.

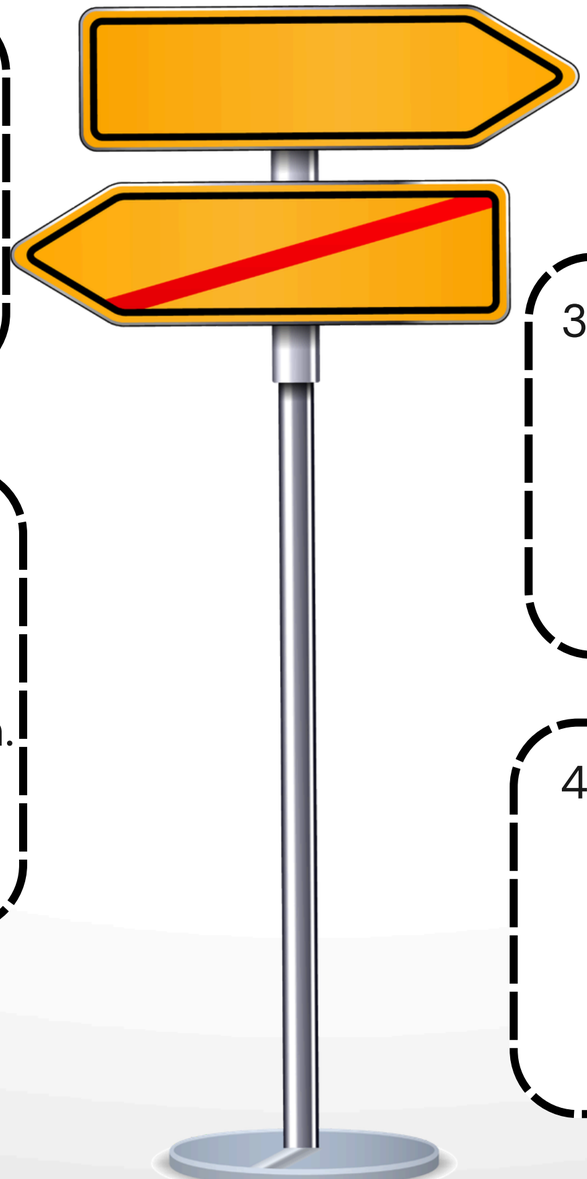
EXISTING SOLUTION:

1. **Fixed-Time Signals:**

   Limitation : Do not adapt their timing to changing traffic demand.

2. **Manual Traffic Control:**

   Limitation : Requires continuous human monitoring and intervention. (microcontroller based timers).



3. **Sensor Based Traffic Control :**

   Limitation : Sensors may provide limited information about vehicle position and movement

4. **Metro City Dynamic Signals:**

   Limitation : Pune, bengaluru dynamic signal has not considered the type of vehicle

OUR IDEA:

A SUMO-based traffic system that dynamically adjusts signal timings and prioritizes emergency vehicles using Python and TraCI.

What is Unique/ Innovative?

1. Traffic-responsive signals:
Green-light duration changes based on simulated traffic conditions rather than relying only on fixed timings.

2. Emergency-vehicle priority:
Includes a rule-based priority mechanism for emergency vehicles in the simulation

3. Timer calculation according to type and count of vehicle:
According to PCU factor given by IRC (Indian road congress)

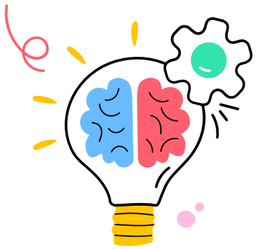
4. Testable simulation environment:
SUMO and TraCI allow traffic scenarios and signal-control decisions to be tested in a simulated road network.



OBJECTIVE:



Build Traffic Simulation



Develop a rule based logic model to manage congestion



Develop Smart Traffic Signals

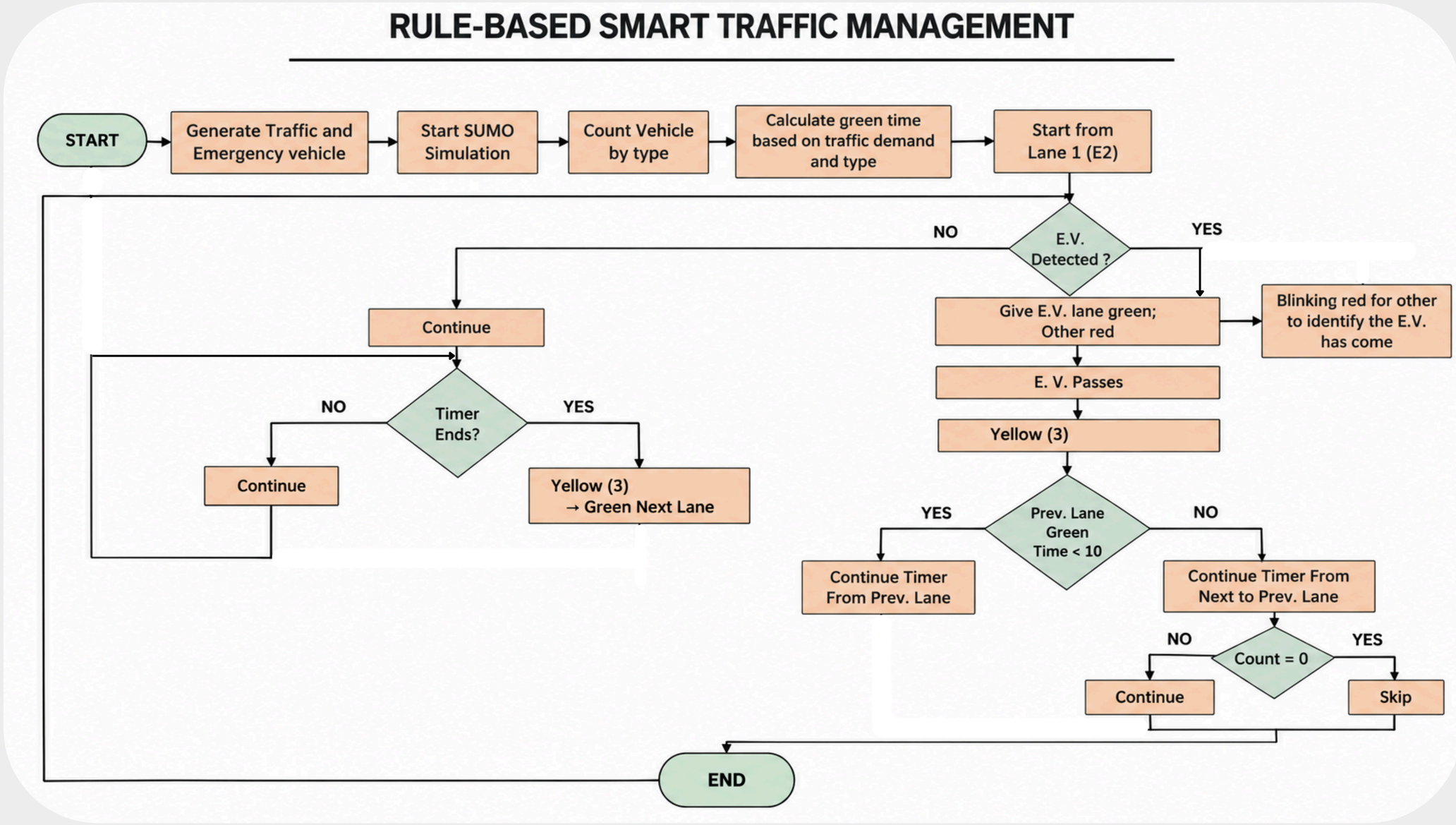


Give Priority to Evergency Vehicles

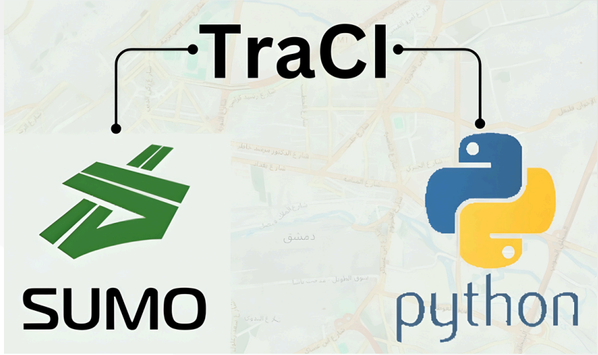


Evaluate System Performance

METHODOLOGY:



TECHNOLOGY ASPECTS





SUMO Simulator :

- **Free & Open-Source**
- **Live AI Connection :** TraCI connects SUMO with Python.
- **Microscopic Simulation :** Tracks individual vehicles and their types.
- **Easy Road Design :** NETEDIT helps create roads, lanes, and intersections easily.

Python :

- **Easy to Use:** Simple syntax to make rule based logic
- **SUMO Integration:** TraCI connects Python with SUMO for real-time monitoring and signal control.

Traci:

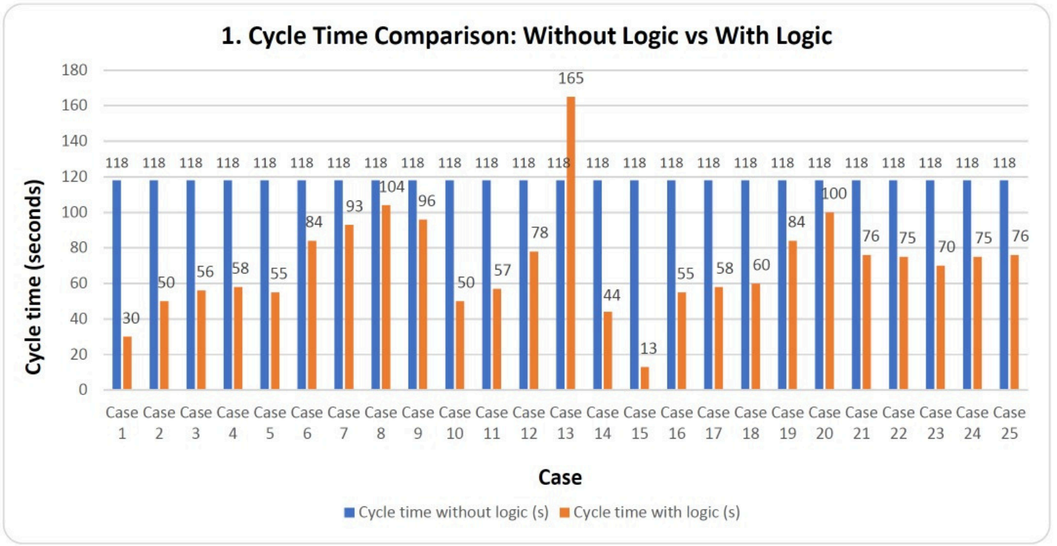
- **Real-Time Communication:** TraCI connects SUMO with Python to receive live vehicle data.
- **Traffic Signal Control:** Allows Python to dynamically control traffic lights.

RESULT & ANALYSIS:

Sr.No	Case	Total vehicle (count)	Car (%)	Bike (%)	Truck (%)	Bus (%)	Van (%)	Time with logic (Sec)	Optimization (%)
1	Less congestion, No E.V.	200	55	25	10	5	5	30	74.58
2	Less congestion, 1 E.V.	200	55	25	10	5	5	50	57.63
3	Less congestion, 2 E.V.	200	55	25	10	5	5	56	52.54
4	Less congestion, 2 E.V. in same lane	200	55	25	10	5	5	58	50.85
5	More congestion, No E.V	300	55	25	10	5	5	55	53.39
6	More congestion, 1 E.V. Green Time > 10 seconds for the previous lane	300	55	25	10	5	5	84	28.81
7	More congestion, 1 E.V. Green Time < 10 seconds for the previous lane	300	55	25	10	5	5	93	21.19
8	More congestion, 2 E.V.	300	55	25	10	5	5	104	11.86
9	More congestion, 2 E.V. in same lane	300	55	25	10	5	5	96	18.64
10	Bike count more, without E.V.	300	25	70	5	5	5	50	57.63
11	Bus count more, without E.V.	300	25	10	5	55	5	57	51.69
12	Truck count more, without E.V.	300	25	10	55	5	5	78	33.90
13	E.V. count more	300	55	25	10	5	5	165	-39.83
14	One lane is empty, and the other lanes are congested	300	55	25	10	5	5	44	62.71
15	Only one lane has vehicles.	300	55	25	10	5	5	13	88.98
16	Less congestion, 1 E.V. Green Time > 10 seconds for the previous lane	300	55	25	10	5	5	55	53.39
17	Less congestion, 1 E.V. Green Time < 10 seconds for the previous lane.	300	55	25	10	5	5	58	50.85
18	Bike count more, with 1 E.V.	300	25	70	5	5	5	60	49.15
19	Bus count more, with 1 E.V.	300	25	10	5	55	5	84	28.81
20	Truck count more, with 1 E.V.	300	25	10	55	5	5	100	15.25
21	2 E.V. at same time (1 Fire truck, 1 Ambulance)	300	55	25	10	5	5	76	35.59
22	2 E.V. at same time (2 Ambulance)	300	55	25	10	5	5	75	36.44
23	2 E.V. at same time (2 Fire truck)	300	55	25	10	5	5	70	40.68
24	4 E.V. at same time (2 Ambuance, 2 Fire truck)	300	35	25	10	5	5	75	36.44
25	4 E.V. at same time (4 Ambulance)	300	55	25	10	5	5	76	35.59
	AVERAGE	-	-	-	-	-	-	**70.48**	**40.27**

Observation Table

$$\text{Optimization (\%)} = \frac{\text{Time without logic} - \text{Time with logic}}{\text{Time without logic}} \times 100$$



Cycle Time Comparison

RESULT & ANALYSIS

Fixed Time Control

118 Sec

Average Waiting Time

Dynamic Time Control

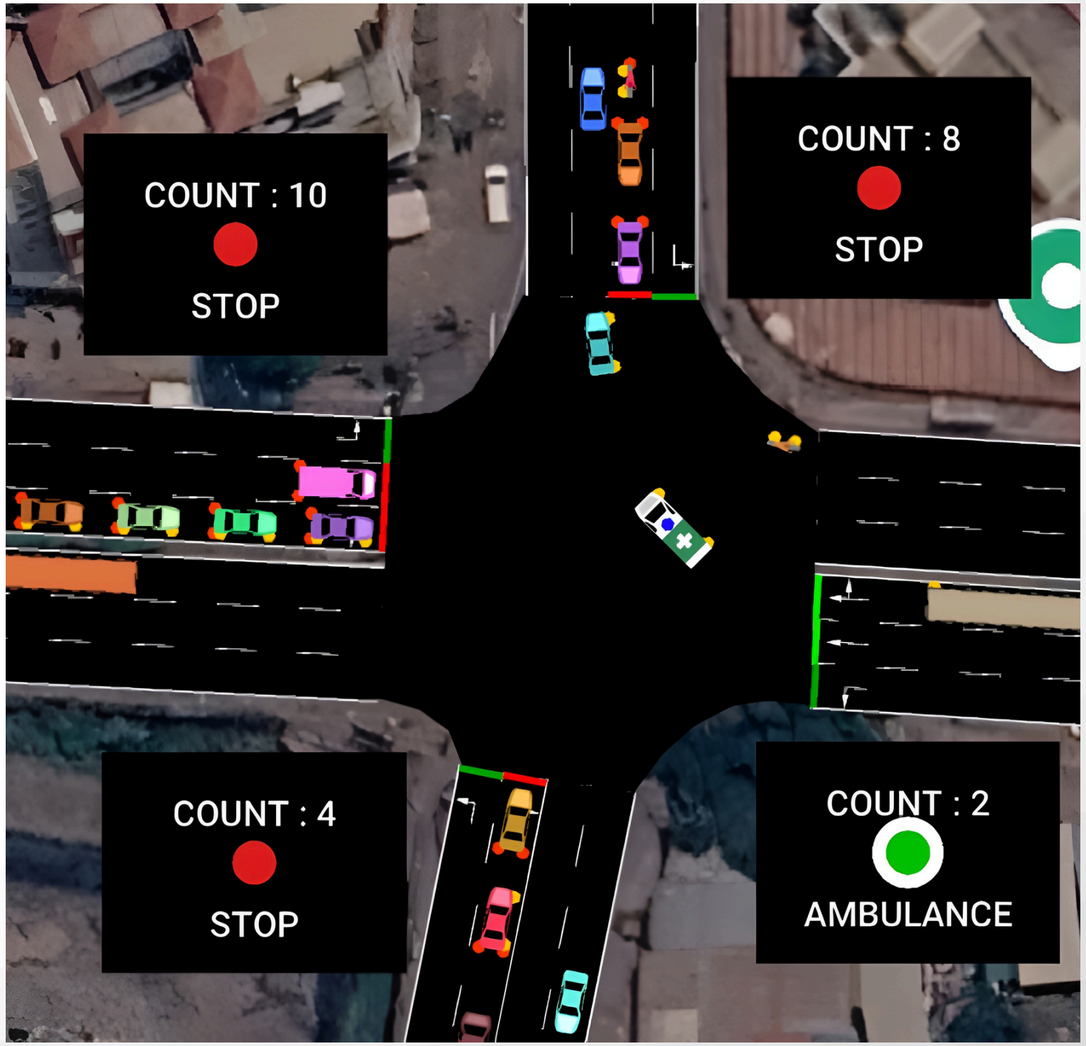
70.48 Sec

Average Waiting Time

Optimization

40.27 %

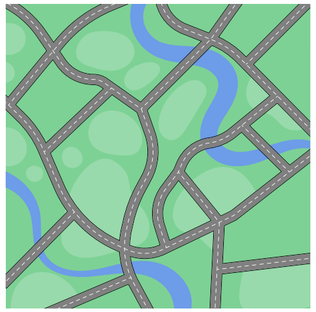
Reduction in Waiting Time



FUTURE SCOPE:

Multiple Junctions

Coordinate signals across multiple intersections.



YOLO Integration

Real-time vehicle detection and counting to calculate signal timers.





Real-World Testing

Implement the system using CCTV cameras and IoT sensors.



AI Prediction

Predict traffic congestion and optimize signal timings.

CONCLUSION:

-: CONCLUSION :-

The Smart Traffic Management System uses SUMO, TraCI, and Python to dynamically adjust traffic signal timings based on vehicle count, vehicle type by rule based control which gives emergency vehicle priority in simulation mode.









Thank You...

Here we are with our full Working Simulation on Smart Traffic Management !!!

Junction iQ